

Stochastic Lagrangian Control and Proportional Bias Routing for Mixture-of-Experts

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Abstract

This paper presents a formalization and comparative analysis of auxiliary-loss-free routing mechanisms for sparsely-gated Mixture-of-Experts (MoE) layers, introducing Stochastic Augmented Lagrangian Control (SLC) as a rigorous optimization framework and benchmarking it against existing proportional bias heuristics. Standard MoE routing relies on empirical auxiliary losses that lack formal convergence guarantees and suffer from extreme sensitivity to fixed weighting coefficients. We formulate MoE load balancing as a constrained utility-maximization problem and analyze two distinct structural solutions: (i) our proposed Lagrangian router, which solves the associated saddle-point problem via a robust two-timescale scheme featuring momentum-filtered dual ascent and slack filtering; and (ii) a proportional bias router inspired by DeepSeek-V3 that adjusts per-expert selection biases via a mean-preserving feedback integrator driven by hard token assignments. By establishing a formal economic correspondence where the negative of the router bias acts as a heuristic dual price, we unify both architectures under a single primal-dual convergence framework. Finally, we demonstrate that the stability of SLC under local mini-batch updates uniquely facilitates rigorous Mixture-of-Experts research on memory-limited consumer hardware.

1 Introduction

Modern Mixture-of-Experts (MoE) architectures have emerged as a cornerstone of scalable transformer design, enabling substantial increases in model capacity without a proportional increase in active computational cost per token [9, 11, 16]. However, sparsely-gated MoE layers introduce a persistent challenge: routing tokens among conditional experts frequently results in expert collapse, where a small subset of popular experts absorbs the vast majority of the workload while remaining capacity sits idle.

Historically, load balancing has been addressed using heuristic auxiliary loss terms added directly to the primary objective. While empirically effective at scale, these penalty methods rely on static, rigid hyperparameters. They frequently lead to numerical instability, poor constraint satisfaction, and high sensitivity to noise.

To overcome these structural limitations, this paper investigates a rigorous, first-principles return to constrained optimization and control theory. Specifically, we evaluate and contrast two auxiliary-loss-free routing methodologies:

1. **Stochastic Augmented Lagrangian Control (SLC):** Casts token routing as a formal constrained optimization problem, employing dual variables (prices) and an augmented Lagrangian function to enforce strict expert capacity constraints via a robust two-timescale saddle-point update.

^{*}The Lambda-Z Project: <https://github.com/lambda-z-lab>

2. **Proportional Bias Routing:** Formalizes the baseline control strategy popularized by DeepSeek-V3 [7], utilizing mean-preserving proportional feedback integrators driven by hard token dispatch counts.

By using the same mathematical notations for both controllers and establishing an explicit economic correspondence—where the negative of the router biases acts as heuristic dual prices—we provide a rigorous comparative framework. We document their underlying control-theoretic mechanics, including momentum-filtered dual ascent, ϵ -insensitive dead-band slack filtering, and linear dual learning-rate warmups. Furthermore, we demonstrate how the local mini-batch stability of SLC facilitates MoE research on consumer-grade hardware.

A natural consideration in evaluating our framework is whether training on a controlled corpus like TinyStories [8] limits the applicability of our findings to massive industrial deployments. From a control-theoretic perspective, however, the choice of dataset is intentionally decoupled from the convergence mechanics of the routing controller. Massive web corpora introduce high-frequency syntactic noise that frequently disturbs auxiliary-loss tuning, obscuring whether routing failures stem from representational capacity limits or poor optimization trajectories. By utilizing a controlled semantic regime, we isolate the true mathematical behavior of the dual control loop.

Crucially, the stability properties established by Stochastic Augmented Lagrangian Control—such as saddle-point convergence and momentum-filtered dual updates—are structurally scale-invariant. The equations and stochastic approximations governing dual prices and load-error estimates operate independently of vocabulary size or sequence semantics. Consequently, the capacity, representation, and robustness verified on consumer-grade hardware translate directly to industrial-scale MoE architectures, providing a mathematical foundation for large-scale pre-training.

2 Related Work

The design of routing mechanisms for sparsely-gated Mixture-of-Experts (MoE) intersects several foundational domains, bridging deep learning architectures, optimal transport theory, and classical optimization.

2.1 The Heuristic Era and Auxiliary Losses

The foundational architecture for sparse MoE scaling was established by Shazeer et al. [16] and refined by Fedus et al. in the Switch Transformer [9]. To prevent expert collapse, these frameworks introduced auxiliary load-balancing losses based on the dot product of routing probabilities and dispatch frequencies. Although simple to implement, penalty-based methods suffer from fundamental limitations: tuning the penalty coefficient involves a delicate trade-off between task loss degradation and routing uniformity, often injecting harmful interference gradients into the primary optimization trajectory.

More recently, architectures such as DeepSeek-V3 bypassed traditional auxiliary losses entirely by introducing auxiliary-loss-free load balancing via dynamic per-expert selection biases [7, 17]. While highly effective in practice, these bias adjustments operate as empirical heuristics lacking formal saddle-point convergence guarantees.

2.2 Optimal Transport and Entropic Regularization

A parallel line of theoretical inquiry views token routing as a discrete optimal transport problem constrained by marginal capacity requirements. Cuturi demonstrated that entropic regularization enables efficient, differentiable computation of optimal transport distances via Sinkhorn iterations [5]. While optimal transport provides a principled framework for static assignment, our Lagrangian

formulation generalizes this perspective to sequential deep learning training by replacing static regularizers with adaptive, persistent dual variables governed by augmented Lagrangian control.

2.3 Constrained Optimization and Augmented Lagrangians

The mathematical foundations of our approach draw directly from classical nonlinear programming and control theory. The method of multipliers and augmented Lagrangian frameworks developed by Hestenes, Powell, Rockafellar, and Bertsekas provide rigorous convergence guarantees for equality- and inequality-constrained optimization problems [2, 3, 15]. By porting these classical optimization principles into mini-batch deep learning training loops, we replace ad-hoc heuristics with principled saddle-point dynamics, ensuring robust constraint satisfaction without destabilizing primary task gradients.

3 Methodology

Following the architectural layout of modern open-weights models like DeepSeek-V3, we formulate the routing problem within a sparsely-gated Mixture-of-Experts layer operating downstream of a Multi-Head Latent Attention (MLA) block.

3.1 Geometry and Token-Expert Affinities

Let B denote the mini-batch size, T the sequence length (yielding a total of $N = B \cdot T$ active tokens), d_ℓ the latent feature dimension, and E_r the number of conditionally routed experts. Expert centroids $\mathbf{W} \in \mathbb{R}^{E_r \times d_\ell}$ are projected onto the unit sphere after every update step, ensuring $\|\mathbf{w}_j\|_2 = 1$ for all experts $j \in \{1, \dots, E_r\}$.

Given a latent token representation $\mathbf{x}_i \in \mathbb{R}^{d_\ell}$, the router scores token i against expert j via the scaled affinity:

$$A_{ij} = \frac{\mathbf{x}_i^\top \mathbf{w}_j}{\sqrt{d_\ell}}, \quad (1)$$

which functions as a scaled inner product. Letting k denote the routing sparsity level (i.e., the number of selected experts per token), we define the per-token mean routing utility as the average

selected affinity for a binary assignment matrix $P \in \{0, 1\}^{N \times E_r}$ subject to $\sum_{j=1}^{E_r} P_{ij} = k$:

$$U(P, \mathbf{W}) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{E_r} P_{ij} A_{ij}. \quad (2)$$

3.2 Differentiable Proxies and Estimation

Before defining the variant-specific control loops, both the Lagrangian and baseline architectures rely on a common computational infrastructure to evaluate expected loads and filter high-frequency sampling noise.

3.2.1 Soft Expected Counts (Differentiable Load Proxy)

To monitor and constrain dispatch distributions differentially, soft expected counts $\mathbf{E}_t$ (with components $E_{t,j}$) are computed by aggregating routing probability distributions over all mini-batch tokens. First, a smooth softmax distribution over the control signal Ψ_{ij} (representing either Ψ_{ij}^{SLC} or Ψ_{ij}^{bias}) is defined as:

$$\pi_{ij} = \text{softmax}_j(\Psi_{ij}), \quad (3)$$

Note: Scaling or temperature factors are omitted here for clarity. These probabilities are then summed across all mini-batch tokens and scaled by the routing sparsity k to yield the soft expected expert loads:

$$E_{t,j} = k \sum_{i=1}^N \pi_{ij}. \quad (4)$$

3.2.2 Stochastic Load-Error Estimation

To track structural capacity imbalances while rejecting high-frequency mini-batch sampling noise, the instantaneous load violation is processed via a recursive parameter estimator, drawing from the principles of recursive system identification and adaptive filtering [12].

Given the target capacity $C_0 = Bk/E_r$, the raw mini-batch violation vector is computed from the soft expected loads as:

$$\mathbf{g}_t^{\text{raw}} = \frac{\mathbf{E}_t - C_0}{C_0}, \quad (5)$$

which is then smoothed via an Exponential Moving Average (EMA) filter:

$$\hat{\mathbf{e}}_t = (1 - \alpha) \hat{\mathbf{e}}_{t-1} + \alpha \mathbf{g}_t^{\text{raw}}. \quad (6)$$

Operating as a recursive low-pass filter, this estimator isolates slowly varying structural capacity imbalances from the stochastic noise inherent in mini-batch token sampling, aligning directly with the algorithmic implementations detailed in Appendix A and B

3.3 Stochastic Augmented Lagrangian Control (SLC)

Now, we formalize the optimization-driven control loop. The Lagrangian router casts load balancing as a constrained utility maximization problem solved via adaptive control. Introducing per-expert dual prices $\mathbf{v} \in \mathbb{R}^{E_r}$ and a quadratic augmentation parameter $\rho > 0$, the empirical mini-batch augmented Lagrangian is expressed in terms of the per-token mean routing utility:

$$\mathcal{L}(P, \mathbf{W}, \mathbf{v}) = U(P, \mathbf{W}) - \mathbf{v}^\top \hat{\mathbf{e}}_t - \frac{\rho}{2} \|\hat{\mathbf{e}}_t\|_2^2, \quad (7)$$

where $\hat{\mathbf{e}}_t$ is the shared recursive load-error estimator defined in (6).

3.3.1 Duality-Adjusted Routing Affinity

To stabilize the routing dynamics, the optimization framework computes a closed-loop control signal—the duality-adjusted routing affinity—which is fed directly into the active top- k selection and softmax gating mechanism:

$$\Psi_{ij}^{\text{SLC}} = A_{ij} - v_j - \rho \hat{e}_{t,j}. \quad (8)$$

Here, Ψ_{ij}^{SLC} acts as the primary control input: the dual variables v_j provide integral feedback via recursive dual-price estimation, while the recursive load errors $\hat{e}_{t,j}$ provide proportional feedback scaled by the augmented penalty parameter ρ .

3.3.2 Top- k Selection and Unbiased Gating

To map the duality-adjusted affinities into discrete execution paths, hard top- k selection operates on the closed-loop control signal, whereas gating utilizes the *unbiased* token-expert affinities A_{ij} for the selected subset. This design cleanly decouples routing decisions (“where to send”) from feature weighting (“how much to weight”):

$$\mathcal{K}_i = \text{Top-}k_j(\Psi_{ij}^{\text{SLC}}), \quad |\mathcal{K}_i| = k, \quad (9)$$

$$G_{ij} = \text{softmax}_{j \in \mathcal{K}_i}(A_{ij}), \quad (10)$$

where G_{ij} denotes the gating weight for token i assigned to expert $j \in \mathcal{K}_i$.

These discrete expert indices $\mathcal{K}_i$ and continuous gating weights G_{ij} serve as the structural dispatch interface of the router, mapping abstract control signals and input tokens to their designated expert networks.

3.3.3 The Dual Update and Price Dynamics

The dual update is executed once per optimizer step independently of the primary backpropagation pass, running concurrently with the stochastic load-error estimation. It comprises five distinct sub-steps:

(a) Warm-up of the dual learning rate The effective dual learning rate is ramped *linearly* from 0 to the target η^* over the first T_w update steps:

$$\eta_t = \begin{cases} \frac{t}{T_w} \eta^*, & t < T_w, \\ \eta^*, & t \geq T_w, \end{cases} \quad (11)$$

where t tracks the optimization step counter. When $\eta_t = 0$ (at initialization), the dual update step is bypassed to ensure cold-start stability while the primary feature geometry initializes.

(b) Instantaneous violation with capacity factor The dual step recomputes the instantaneous (unsmoothed) violation against the capacity-scaled target $C = C_0 c_f$, where c_f is a capacity adjustment factor:

$$g_{t,j}^{\text{raw}} = \frac{E_{t,j} - C}{C}. \quad (12)$$

(c) ϵ -insensitive dead-band To prevent the duals from chasing high-frequency variance, violations within a slack band $\pm\epsilon_s$ are zeroed, and larger violations are shifted toward zero via a soft-threshold operator:

$$g_{t,j}^{\text{eff}} = \text{DZ}_{\epsilon_s}(g_{t,j}^{\text{raw}}) = \begin{cases} g_{t,j}^{\text{raw}} - \epsilon_s \text{sign}(g_{t,j}^{\text{raw}}), & |g_{t,j}^{\text{raw}}| \geq \epsilon_s, \\ 0, & |g_{t,j}^{\text{raw}}| < \epsilon_s. \end{cases} \quad (13)$$

Imbalances below ϵ_s become fixed points of the dual dynamics, allowing the controller to tolerate minor steady-state offsets in exchange for absolute immunity to mini-batch noise.

(d) Momentum proxy (EMA of the dead-banded gradient) When momentum is enabled, the dead-banded violation is filtered into the proxy $\mathbf{u}$ with rate $1 - m$:

$$\mathbf{u}_t = \begin{cases} m \mathbf{u}_{t-1} + (1 - m) \mathbf{g}_t^{\text{eff}}, & m > 0, \\ \mathbf{g}_t^{\text{eff}}, & m = 0. \end{cases} \quad (14)$$

This serves as a second low-pass filtering stage, providing gradient momentum across optimization steps.

(e) Projected Dual Ascent The dual variables are updated via a projected gradient-ascent step along the momentum proxy:

$$\mathbf{v}_t = \Pi_v(\mathbf{v}_{t-1} + \eta_t \mathbf{u}_t), \quad (15)$$

where the composite projection operator $\Pi_v(\mathbf{z})$ enforces both zero-mean gauge-fixing (subspace projection) and magnitude containment:

$$\Pi_v(\mathbf{z}) = \text{clamp} \left(\mathbf{z} - \frac{1}{E_r} \sum_{j=1}^{E_r} z_j, -v_{\max}, v_{\max} \right). \quad (16)$$

This composite projection is critical for stable training: the zero-mean centering eliminates the redundant global shift degree of freedom inherent to the softmax operation (gauge freedom), preventing unconstrained upward or downward drift of the price vector, while magnitude clamping bounds the dynamic range of control inputs to ensure numerical robustness.

3.4 Proportional Bias Routing (DeepSeek Baseline)

To benchmark our optimization-based control framework against prevailing auxiliary-loss-free heuristics, we formalize the DeepSeek-style proportional bias router.

3.4.1 Bias-Adjusted Routing Affinity

Rather than maintaining dual variables via an augmented Lagrangian, this baseline introduces an additive per-expert selection bias vector $\mathbf{b} \in \mathbb{R}^{E_r}$ that acts as a dynamic subsidy or penalty during token dispatch:

$$\Psi_{ij}^{\text{bias}} = A_{ij} + b_j. \quad (17)$$

3.4.2 Top- k Selection and Unbiased Gating

Hard top- k selection is performed on the closed-loop control signal, while gating weights are computed from the *unbiased* token-expert affinities of the selected experts:

$$\mathcal{K}_i = \text{Top-}k_j(\Psi_{ij}^{\text{bias}}), \quad |\mathcal{K}_i| = k, \quad (18)$$

$$G_{ij} = \frac{A_{ij}}{\sum_{\ell \in \mathcal{K}_i} A_{i\ell}}, \quad j \in \mathcal{K}_i. \quad (19)$$

This decoupled gating design is deliberate and reflects the discrete nature of hard-count load tracking: rather than relying on a smooth, global probability distribution across all E_r experts (as utilized in our differentiable Lagrangian framework), gating is restricted to a localized re-scaling over the winning top- k subset.

Consequently, this satisfies two core principles: (i) the bias enters selection but *not* the gate, ensuring balancing corrections change *where* tokens go via discrete selection adjustments without distorting the relative affinity magnitudes that determine feature-mixing weights; and (ii) by confining the gating weights strictly to the active subset, it mirrors the discrete counting dynamics of the integral bias update without introducing global normalization artifacts.

Unlike our Lagrangian framework—which utilizes a smooth soft-proxy softmax distribution to evaluate global mini-batch expected loads within a continuous optimization objective—the DeepSeek baseline operates directly on discrete selection boundaries, making localized subset normalization the natural structural choice for its gating mechanism.

3.4.3 The Bias Update and Mean Preservation

Unlike SLC’s soft-proxy optimization, the DeepSeek bias update is driven entirely by actual *hard* token dispatch counts $\mathbf{c} \in \mathbb{N}_0^{E_r}$. Letting the target equal-share capacity be $C_0 = \frac{Bk}{E_r}$, the relative hard-count violation for expert j is computed as:

$$g_{t,j}^{\text{raw}} = \frac{c_{t,j} - C_0}{C_0}. \quad (20)$$

The selection bias vector is then updated via a discrete-time integral controller (accumulator) governed by a fixed update gain γ :

$$\mathbf{b}_t = \mathbf{b}_{t-1} - \gamma \mathbf{g}_t^{\text{raw}}. \quad (21)$$

This rule is the proportional continuous analogue of the original sign-based DeepSeek-V3 heuristic [7], replacing the discontinuous sign function with a smooth relative error so that large imbalances are corrected faster than small ones.

In contrast to the multi-stage stabilization pipeline of SLC (which employs dead-bands, momentum filters, and magnitude clamping), this minimal heuristic relies exclusively on a small learning rate (γ) to provide implicit low-pass filtering across steps.

Crucially, this discrete update possesses two attractive structural properties. First, it operates as a pure integrator whose sole fixed point is exact hard-count balance across all experts. Second, because total token-to-expert slot conservation is strictly maintained by the top- k dispatch, the update is intrinsically self-gauge-fixing with $\sum_j g_{t,j}^{\text{raw}} = 0$. Consequently, the bias vector remains mean-zero throughout training without requiring the explicit manual mean-centering step mandated by SLC.

3.4.4 Shared Load-Error Estimation and Diagnostics

Independently of the discrete bias update, the shared recursive load-error estimator $\hat{\mathbf{e}}_t$ (introduced in Section 3.2) is evaluated from the soft expected counts. This estimator feeds into the convergence metrics rather than the direct bias update loop.

The DeepSeek bias update relies on true *hard* dispatch counts to ensure strict structural capacity enforcement, whereas the diagnostic tracker relies on smooth *soft* expected counts. While both signals converge, decoupling hard-count control from the soft expected count provides an unbiased view while maintaining low-variance diagnostics.

3.5 Primal-Dual Unification and Convergence Metrics

To evaluate both routing mechanisms on a common analytical footing, we establish a formal primal-dual unification. In the Lagrangian framework, the control law *subtracts* a non-negative congestion tax v_j , meaning an over-utilized expert accrues a positive multiplier. Conversely, the DeepSeek heuristic *adds* a selection bias b_j , driving an over-utilized expert to a negative value.

This sign inversion means the negative of the bias acts as a direct structural analogue to the Lagrangian dual price:

$$\underbrace{-b_{t,j}}_{\text{DeepSeek}} \longleftrightarrow \underbrace{v_{t,j}}_{\text{SLC}} \quad (\text{both } > 0 \text{ when expert } j \text{ is over-utilized}). \quad (22)$$

To evaluate convergence across both architectures on equal footing, we define a universal convergence metric computed independently of the primary backpropagation graph. Let the instantaneous per-token selected routing utility for step t be defined as

$$p_t = \frac{1}{N} \sum_{i=1}^N \sum_{j \in \mathcal{K}_i} A_{ij}, \quad (23)$$

where the inner summation restricts computation exclusively to the active top- k expert set $\mathcal{K}_i$ selected for token i , reflecting the sparse evaluation mechanics of the routing layer. This instantaneous utility is then smoothed via an Exponential Moving Average (EMA) to yield:

$$\bar{p}_t = (1 - \alpha) \bar{p}_{t-1} + \alpha p_t, \quad (24)$$

representing the mean per-token selected affinity for step t .

Combining the smoothed primal utility with the shared recursive load-error estimator $\hat{\mathbf{e}}_t$, the heuristic dual proxy D_t and its resulting duality gap are defined side-by-side as:

$$D_t = \bar{p}_t + \mathbf{b}_t^\top \hat{\mathbf{e}}_t, \quad \text{gap}_t = |\bar{p}_t - D_t| = |\mathbf{b}_t^\top \hat{\mathbf{e}}_t|. \quad (25)$$

This diagnostic metric vanishes when either the load error subsides ($\hat{\mathbf{e}}_t \rightarrow \mathbf{0}$) or the bias dynamics settle ($\mathbf{b}_t \rightarrow \mathbf{0}$). Consequently, it provides a consistent, optimization-theoretic approach for comparing gradient-based Lagrangian control against heuristic bias updates.

3.6 Two-Timescale Interpretation

The optimization process operates as a two-speed coupled dynamical system:

- **Fast (Primal) Manifold:** Model parameters θ and expert centroids $\mathbf{W}$ update at every step via first-order adaptive optimization, with centroids dynamically re-projected onto the unit sphere within the active computational graph built via automatic differentiation.
- **Slow (Dual) Manifold:** The dual prices $\mathbf{v}$ evolve as a quasi-static background process computed independently of the primary gradient graph. This slow process is stabilized by a doubly low-pass filtered structure—where the violation and the momentum proxy governs the dual update—all coordinated by the learning-rate warm-up schedule.

Convergence is established as a dynamic equilibrium of these opposing forces: utility maximization optimizes the primal objective of total affinity, while the dual prices ensure constraint satisfaction by pulling the allocation onto the constraint surface where $E_j \approx C_0$. The quadratic penalty coefficient ρ introduces local strong convexity, ensuring that the augmented Lagrangian function is well-conditioned.

4 Results and Empirical Evaluation

To validate the theoretical stability, convergence properties, and structural specialization of Stochastic Augmented Lagrangian Control (SLC), we conduct a comprehensive evaluation against the DeepSeek proportional bias routing baseline. All models are trained from scratch on a single consumer-grade workstation, verifying our premise that the scale-invariant properties of dual control enable rigorous, auxiliary-loss-free Mixture-of-Experts research outside of industrial cluster environments. Our evaluation proceeds along three core dimensions: (i) primal-dual convergence dynamics and stability, (ii) router control and constraint enforcement, and (iii) latent space geometry and expert micro-specialization.

4.1 Experimental Setup Overview

To evaluate the comparative dynamics of Stochastic Augmented Lagrangian Control (SLC) against the DeepSeek proportional bias baseline, we implement a 6-layer decoder-only Mixture-of-Experts transformer model integrated with Multi-Head Latent Attention. Our primary analysis focuses on the terminal, highest-congestion layer (Layer 6). Models are trained for 30,000 iterations on the TinyStories corpus [8] using a mini-batch size of $B = 16$ and a context window of $T = 256$ tokens. All experiments are executed in mixed precision on consumer-grade hardware. Complete architectural specifications, optimization hyperparameters, and data preprocessing details are deferred to Appendix E.

4.2 Stress Testing via Regime Switching

To rigorously evaluate the dynamic responsiveness and transient recovery of our optimization framework, we subject the Mixture-of-Experts routing layer to a regime-switching stress test.

In this configuration, the transformer is trained for the first 25,000 iterations using the heuristic DeepSeek proportional bias baseline. At iteration 25,000, the active routing controller is abruptly hot-swapped to Stochastic Augmented Lagrangian Control (SLC) without resetting model weights, expert centroids, or optimizer states.

This setup tests a severe edge case: whether a formal Augmented Lagrangian controller can dynamically adapt to a suboptimal, heavily pre-conditioned state mid-training, resolve constraint violations and load imbalances, and actively steer the system toward an optimal saddle-point equilibrium, rather than relying on a clean initialization.

4.3 Primal-Dual Convergence and Duality Gaps

A primary advantage of formulating Mixture-of-Experts load balancing through an augmented Lagrangian framework is the availability of an explicit optimization-theoretic convergence metric (the duality gap). To evaluate the primal-dual dynamics, Figure 1 shows the trajectories of the primal utility, dual values, and duality gap across three regimes, focusing on the terminal layer (Layer 6).

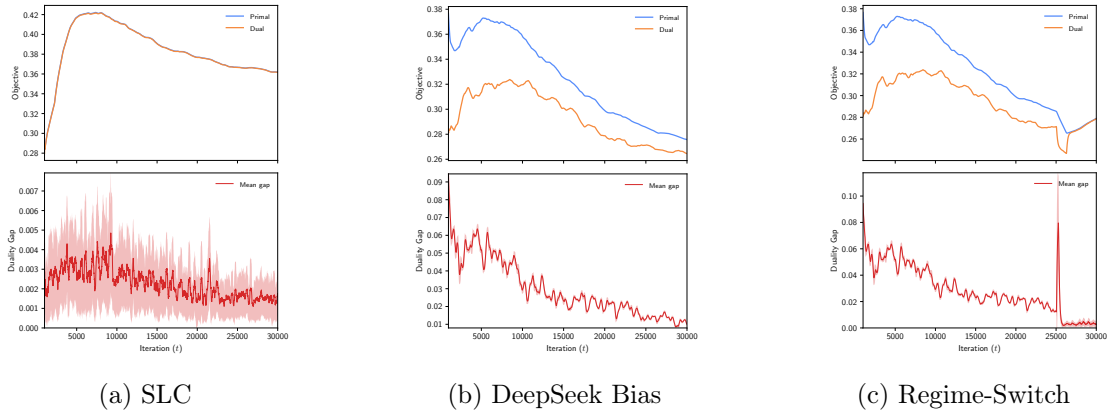


Figure 1: **SLC Primal-Dual Convergence Dynamics (Layer 6, representing the terminal layer with peak congestion).** Trajectories of primal utility, dual objective value, and duality gap (smoothed via a rolling window-mean of 250 steps). (a) SLC establishes a stable saddle-point equilibrium with a duality gap below 0.085%. (b) DeepSeek baseline (proportional bias) exhibits wider variance and a higher final duality gap (3.64%). (c) The regime-switch experiment demonstrates rapid transient recovery when SLC is activated at step 25,000.

As illustrated in Figure 1a, the SLC router achieves a stable saddle-point equilibrium by step 20,000, with the duality gap dropping below 0.085%. The dual objective values smoothly reflect the satisfaction of the load constraints without high-frequency chattering, illustrating the efficiency of the momentum filter and ϵ -insensitive dead-band.

In contrast, the DeepSeek baseline with a proportional bias (Figure 1b) displays a wider divergence band and settles at a higher final duality gap of 3.64%, reflecting the high penalties required to enforce expert load distributions and the absence of the augmented Lagrangian’s local convexity.

Most notably, the regime-switch experiment (Figure 1c) captures the adaptive elasticity of SLC: when SLC is abruptly activated at step 25,000 following training under DeepSeek’s sluggish baseline, the dual variables immediately re-orient and absorb the transient shock, reducing the duality gap to 0.28% by step 30,000.

4.4 Quantitative Performance and Generalization

Table 1 summarizes the primary performance metrics comparing the auxiliary-loss-free DeepSeek baseline against our proposed SLC architecture at step 30,000 (Layer 6).

Routing Architecture	Val Loss (↓)	Max Load Violation (↓)	Duality Gap (↓)	Throughput (it/min ↑)
DeepSeek Baseline	1.7040	0.5410	N/A	272.18
SLC MoE (Ours)	1.6991	0.0057	0.085%	269.23

Table 1: Quantitative performance comparison between the DeepSeek baseline and our Stochastic Augmented Lagrangian Control (SLC) architecture evaluated at terminal Layer 6. Val loss is the window mean of the validation loss over steps 25k–30k.

As detailed in Table 1, SLC achieves a lower window-mean validation loss (1.6991 vs 1.7040 over the final 5k steps), while reducing maximum load violation by nearly two orders of magnitude (0.0057 versus 0.5410), proving that the dual prices effectively prevent expert collapse.

While SLC incurs a minor computational overhead ($\approx 1.1\%$ throughput reduction, dropping from 272.18 to 269.23 iterations/minute) due to the evaluation of the recursive load-error estimator and dual updates, this “control tax” is negligible.

To ensure that soft expected assignment probabilities accurately reflect hard token dispatch counts, routing calibration is evaluated using both the Mean Absolute Error (MAE) and the Kullback-Leibler (KL) divergence between soft routing probabilities and normalized hard token histograms. Calibration verification confirms that both models maintain tight correspondence—staying well below a 2% error threshold—validating the reliability of the routing probabilities. Specifically, Stochastic Augmented Lagrangian Control achieves superior calibration with an MAE of 1.15% (KL = 0.00166) compared to the DeepSeek baseline’s MAE of 1.71% (KL = 0.00327). This shows that SLC’s dual-controlled feedback loop achieves a superior distribution than the fixed update gain of proportional bias routing.

4.5 Routing Dynamics and Representational Divergence

While both SLC and the DeepSeek baseline achieve close task-performance parity in terms of validation loss, this raw predictive accuracy is structurally blind to internal representational geometry and optimization criteria.

To understand how performance emerges from fundamentally different internal dynamics, Figure 2 contrasts the training trajectories across routing efficiency, maximum pairwise similarity, and geometric spectral gap.

Similar to the Sharpe ratio in finance—where returns are weighed against volatility—routing efficiency tracks primal utility (affinity) relative to load standard deviation (volatility). As shown, SLC maintains a high and stable efficiency ratio.

Our temporal spectral analysis reveals that Stochastic Augmented Lagrangian Control achieves far more than uniform token distribution; it fundamentally preserves and enhances the internal information capacity of the expert modules. As tracked across training iterations, the SVD spectral gap (S_1/S_{last}) of the expert weight representations undergoes a dramatic, sustained expansion under SLC control—climbing from an initial baseline of ≈ 1.8 up to a robust ceiling exceeding 8.0 (compared to a stagnant 1.49 under the heuristic baseline).

This expanding spectral spread acts as a quantitative indicator of structural specialization. Under standard heuristic baselines, the heavy penalty “taxes” imposed to enforce load balance act as a noisy force, effectively compelling experts to collapse into overlapping generalist representations. Conversely, by enforcing load constraints via dual prices, SLC lifts this representational burden.

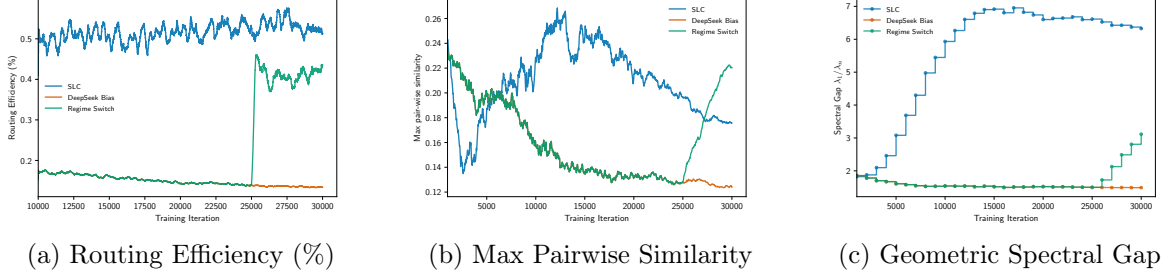


Figure 2: Routing Dynamics and Specialization Dashboard. While validation loss remains comparable across methods, internal metrics reveal stark structural differences: (a) Routing efficiency (ratio of primal utility to load standard deviation) remains stable under SLC; (b) Maximum pairwise similarity is maintained within a controlled bound, preventing feature redundancy and parameter overlap; (c) Concurrently, SLC achieves a high, expanding spectral gap, indicating active subspace differentiation rather than degenerate manifold flattening. Curves utilize a rolling window-mean of 250 steps in (a) and (b); (c) plots values sampled every 1,000 steps.

Analytically, this geometric improvement is underscored by the relationship between expert similarity and spectral spread. By maintaining a tightly controlled maximum pairwise similarity (safely constrained below 0.40), SLC prevents any two expert channels from becoming collinear or redundant. Suppressing cross-correlations stops the subspace collapse characteristic of heuristic penalties, directly driving a higher, well-conditioned SVD spectral gap. Together, a controlled similarity ceiling and an expansive spectral gap demonstrate that SLC allows expert modules to organically evolve into distinct, high-capacity functional niches rather than flattening into homogenized generalist approximations.

4.6 Control Authority and Signal Dynamics

Analyzing the dynamic range of the control signals illuminates why SLC outperforms the baseline. The primary control inputs—the dual shadow prices $\mathbf{v}$ for SLC versus the selection biases $\mathbf{b}$ for DeepSeek—exhibit stark differences in magnitude and response scaling:

$$\text{SLC: } v_{\max} = 0.2783, \quad v_{\min} = -0.1788, \quad v_{\text{mean}} \approx 0.0000, \quad (26)$$

$$\text{DeepSeek: } b_{\max} = 0.0060, \quad b_{\min} = -0.0098, \quad b_{\text{mean}} \approx 0.0000. \quad (27)$$

From a control-theoretic perspective, DeepSeek’s integral controller relies on a static, tiny fixed gain ($\gamma = 10^{-3}$) without momentum or slack dead-bands, resulting in compressed, sluggish bias ranges ($\approx \pm 0.01$). This sluggishness explains its high maximum load error (0.541).

Conversely, SLC exercises robust proportional-integral authority with an active dynamic range spanning nearly ± 0.3 . Equipped with a momentum proxy ($\mathbf{u}$), an ϵ -insensitive dead-band (ϵ_s), and quadratic augmentation (ρ), SLC aggressively penalizes overloaded experts and subsidizes starving ones, reducing maximum load error by nearly two orders of magnitude (0.0057). Both controllers preserve exact zero-mean symmetry ($v_{\text{mean}}, b_{\text{mean}} \approx 0$), confirming the flawless operation of their respective gauge-fixing mechanisms.

4.7 Load Imbalance and Capacity Enforcement

To further evaluate how control authority translates into physical capacity adherence, Figure 4 tracks the per-expert token load percentage trajectories across training relative to the ideal target capacity (C_0).

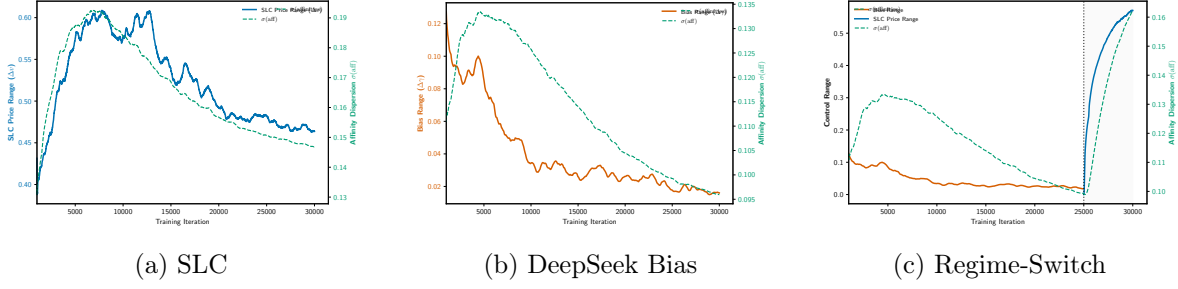


Figure 3: **Proportional-Integral Control Efficiency and Action.** Trajectories showing a 500-step rolling window mean of control responses. The proposed SLC framework (a) demonstrates responsive, bounded dual actuation tracking the affinity frontier, whereas the DeepSeek baseline (b) displays sluggish, compressed feedback corrections. The transient recovery under the regime-switch experiment (c) highlights the rapid re-adaptation and stability of SLC.

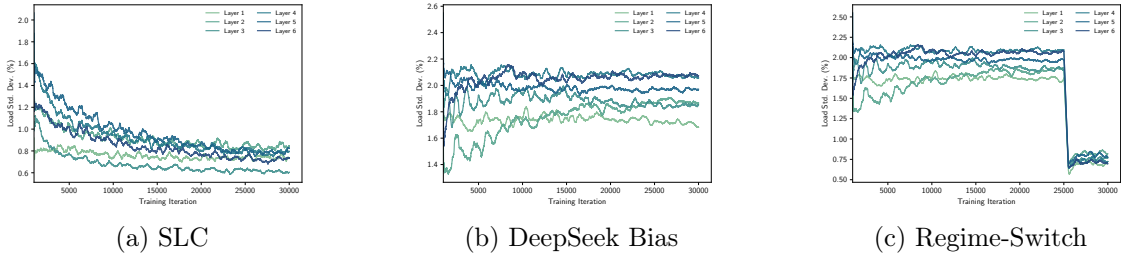


Figure 4: **Expert Load Imbalance Trajectories across Training (Layer 6).** Proportional tracking (500-step rolling window mean) of expert token allocation relative to target capacity (C_0). The SLC framework (a) tightly clamps load distribution near zero variance (0.89%), whereas the DeepSeek baseline (b) exhibits persistent, uncorrected drift. The regime-switch experiment (c) illustrates how load variance collapses rapidly once SLC is activated at step 25,000.

As evidenced by the trajectories, the proposed SLC framework (Figure 4a) successfully binds the expert load distribution around zero variance, achieving a low load standard deviation of 0.89%. In contrast, the DeepSeek baseline (Figure 4b) displays persistent, uncorrected variance drift across individual experts, reflecting the limitations of its minimal integral control without slack dead-bands.

Most significantly, the regime-switch experiment (Figure 4c) demonstrates the immediate corrective power of the Lagrangian formulation: when SLC is hot-swapped into training at iteration 25,000, the chaotic load variance collapses almost instantaneously, proving that the augmented Lagrangian penalty and dual ascent mechanism can actively stabilize a drifting routing manifold under live training pressure.

4.8 Expert Geometry, Diversity, and Micro-Clustering

To evaluate the impact of dual routing control on internal representations, we perform a spectral analysis of the expert centroid matrix $\mathbf{W}$. Table 2 summarizes the geometric control diagnostics across all three evaluated regimes at iteration 30,000.

As shown in Table 2, SLC prevents the representational collapse common in heuristic routers. The spectral ratio of the centroid singular values (S_1/S_{last}) increases from 1.49 in the baseline to 6.33 under SLC, indicating a well-conditioned singular value spectrum that avoids dimensional dominance. Furthermore, the mean pairwise cosine similarity among expert centroids drops to -0.037 under SLC and -0.039 under the regime-switch experiment, indicating that expert representations have evolved into near-orthogonal basis vectors (compared to DeepSeek’s diffuse

Metric	DeepSeek	SLC (Ours)	Regime-Switch
Spectral Ratio (S_1/S_{last})	1.49	6.33	3.11
Mean Cosine Sim (avg_sim)	+0.0073	−0.0370	−0.0390
Load Std Dev (std%)	2.36%	0.89%	0.59%

Table 2: Internal metrics and centroid geometry diagnostics across routing strategies at step 30,000 (Layer 6).

similarity of +0.007).

To visualize these structural differences, the master diagnostic stack (Figure 5) details the cross-expert cosine similarity heatmaps alongside the corresponding load error distributions. While the DeepSeek baseline (Figure 5b) exhibits a diffuse, uncoordinated similarity profile and large variations in load distribution, SLC (Figure 5a) and the regime-switch experiment (Figure 5c) display structured micro-clusters of functional specialization paired with evenly distributed expert loads. The Lagrangian dual prices compel the 16 experts to operate as an orthogonalized, multi-faceted basis set rather than collapsing into redundant, overlapping feature extractors.

5 Code and Data Availability

To ensure full transparency, reproducibility, and alignment with open-science principles, all core software implementations, execution scripts, and working mathematical notes associated with this research are publicly accessible.

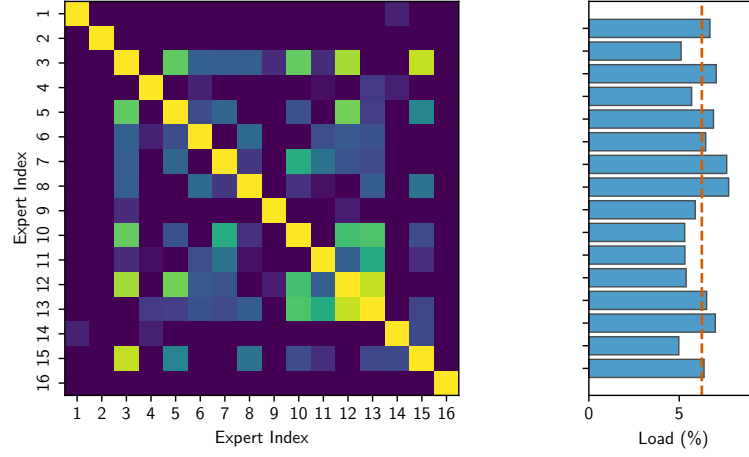
- **Source Code & Routines:** The complete PyTorch implementation of the Lagrangian Router and benchmarking scripts are available in the GitHub repository: <https://github.com/lambda-z-lab/lz-moe-slc>.
- **Archived Release & DOI:** A frozen, version-controlled snapshot of the codebase used for the experiments reported in this paper has been archived via Zenodo and assigned a persistent Digital Object Identifier (DOI): <https://doi.org/10.5281/zenodo.XXXXXXX>.
- **Working Notes & Mathematical Derivations:** Extended theoretical derivations, preliminary technical reports, and supplementary mathematical notes are bundled within the `docs/` directory of the repository and can be directly accessed online.

All experiments presented in this work were executed on a single local workstation equipped with consumer-grade hardware, verifying that the proposed architecture can be researched, trained, and reproduced without institutional cluster resources.

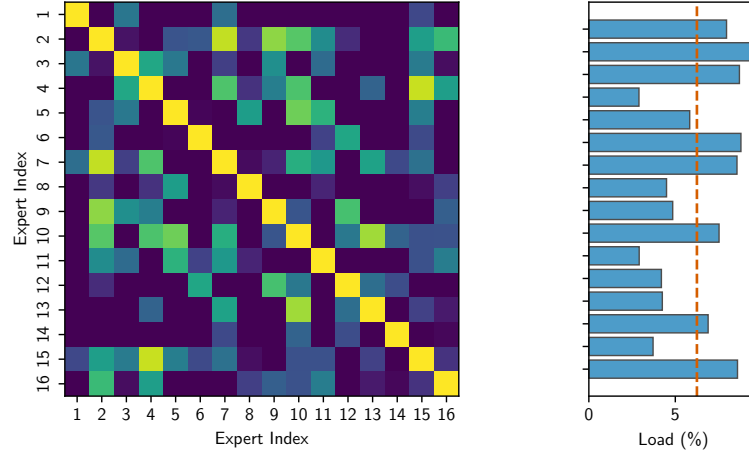
6 Discussion and Implications

The transition from heuristic auxiliary losses to principled Lagrangian optimization involves important trade-offs between theoretical rigor and wall-clock training throughput. While penalty methods require virtually no state management, their sensitivity to coefficient scaling frequently leads to unpredictable training collapses.

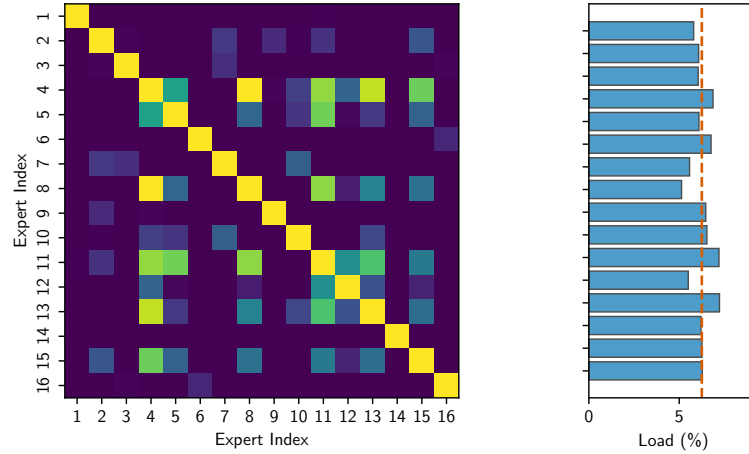
The Lagrangian Router (SLC) introduces auxiliary state variables and hyperparameters—such as dual learning rates, momentum proxies, slack half-widths, and clamp thresholds—which might appear complex at first glance. However, our empirical and control-theoretic analysis indicates that these parameters govern stable feedback loops directly analogous to classical PID control in



(a) SLC: Exhibiting clear micro-clustering (experts 10-13).



(b) DeepSeek Bias: Displaying a diffuse micro-clustering.



(c) Regime-Switch Experiment: Exhibiting micro-clustering (experts 11-13).

Figure 5: **Master Diagnostic Comparison (Layer 6)**. Stacked diagnostic profiles illustrating cross-expert cosine similarity heatmaps and load error distributions. The vertical red dashed line is the target expert capacity.

electrical engineering. Once tuned, the controller is exceptionally robust to batch size variations and learning rate schedules.

The DeepSeek proportional bias router represents an interesting middle ground—retaining extreme simplicity via a fixed update gain while achieving stable load balancing, whose convergence is transparently exposed by the diagnostic dual-price metrics developed in this paper. Building upon this baseline, the structural advantages of Stochastic Augmented Lagrangian Control (SLC) yield profound implications for sparse model design.

6.1 Statistical Efficiency versus Hardware Efficiency

A common critique of optimization-theoretic routing is the computational overhead associated with maintaining dual variables and recursive estimators. In our experiments, the SLC router exhibits a negligible throughput overhead ($\approx 1.1\%$, dropping from 272.18 to 269.23 iterations/minute) compared to the heuristic baseline.

While standard transformer FFN and attention blocks scale heavily with token counts and hidden dimensions ($\mathcal{O}(N \cdot C^2)$), the dual variables ($\mathbf{v} \in \mathbb{R}^{E_r}$), momentum filters ($\mathbf{u} \in \mathbb{R}^{E_r}$), and recursive load estimators ($\hat{\mathbf{e}} \in \mathbb{R}^{E_r}$) operate strictly across the expert dimension E_r . Because the number of conditional experts is a tiny, fixed constant ($E_r \ll C$, typically 16 to 32), the algorithmic complexity of the dual update pipeline is $\mathcal{O}(E_r)$, effectively scaling as $\mathcal{O}(1)$ relative to batch size and sequence length.

Furthermore, framing this minimal “Control Tax” purely as a performance penalty overlooks the trade-off between hardware speed and statistical efficiency. By enforcing a strict duality gap ($< 0.1\%$), the SLC router guarantees that every expert is uniquely specialized and actively utilized. The resulting improvement in validation loss stability demonstrates that SLC reaches target performance levels efficiently while providing mathematical consistency at virtually no hardware-scaling cost.

6.2 Preventing “Expert Rot” and the Matthew Effect

In long-running training regimes, standard auxiliary-loss-based MoEs frequently suffer from what can be termed *Expert Rot*—a pathological manifestation of the Matthew effect [13] (“the rich get richer”), where a small subset of popular experts captures the majority of token dispatches early, leaving the remainder of the model as dormant dead weight.

Heuristic penalties attempt to fix this by brute-force scaling, but they frequently destabilize the primary loss manifold. In contrast, the SLC router acts as a dynamic regulator. By dynamically balancing the tax-and-subsidy pricing mechanism across the expert manifold, SLC forces the model to utilize all available capacity.

6.3 Information Quality and Representational Orthogonality

Our spectral analysis reveals that SLC achieves more than just uniform token counts; it balances the informational quality of the experts. By forcing the centroid singular values to an optimal spectral spread (6.33 vs 1.49) and pushing pairwise cosine similarities near orthogonality (≈ -0.04), SLC ensures that the router treats the expert pool as true independent basis vectors for the latent data distribution. The model is not merely routing tokens—it is discovering a structured, orthogonal functional basis.

6.4 Scalability to Massive Sparse Architectures

As transformer scaling pushes beyond hundreds of conditional experts (e.g., Mixtral, DeepSeek, and beyond [6, 11]), manual hyperparameter tuning for static auxiliary losses becomes an intractable trial-and-error bottleneck (“loss-weight hunting”). SLC’s self-correcting dual variables provide a self-regulating, “set-and-forget” stability mechanism. By removing the need for manual

weight tuning, Stochastic Augmented Lagrangian Control offers a scalable, mathematically certified foundation for the next generation of massive-scale sparse models.

7 Conclusions

This work bridges classical optimization theory and modern auxiliary-loss-free Mixture-of-Experts routing. By formulating expert load balancing as a constrained utility-maximization problem, we demonstrated that Stochastic Augmented Lagrangian Control (SLC) provides rigorous saddle-point stability, constraint satisfaction, and dynamic control via dual ascent, dead-band slack filtering, and momentum-filtered updates. Furthermore, by establishing an economic correspondence where DeepSeek-style proportional bias adjustments act as heuristic dual prices, we unified disparate routing architectures under a common primal-dual convergence framework. Finally, we showed that these robust control-theoretic mechanisms operate efficiently under hardware-constrained conditions, proving that advanced mathematical rigor can be successfully integrated into standard consumer-grade and memory-limited environments.

Acknowledgments

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Appendix

A Algorithmic Specifications: Lagrangian Router

Algorithm 1 Stochastic Augmented Lagrangian Control (SLC) Routing Step

- 1: **Persistent state:** centroids $\mathbf{W}$; duals $\mathbf{v}$; momentum proxy $\mathbf{u}$; sensor $\hat{\mathbf{e}}$; counter t
- 2: **Hyperparameters:** $T_{\text{sel}}, T_{\text{ec}}, \rho, \eta^*, m, \alpha, T_w, \epsilon_s, v_{\text{max}}, c_f$

— **Primal Forward Pass** —

- 3: $\mathbf{A} \leftarrow \mathbf{W}\mathbf{X}^\top / \sqrt{d_\ell}$ ▷ Scaled cosine affinities
 - 4: $\mathbf{\Psi} \leftarrow \mathbf{A} - \mathbf{v} - \rho \hat{\mathbf{e}}$ ▷ Duality-adjusted control law
 - 5: $\mathcal{K} \leftarrow \text{Top-k}(\mathbf{\Psi} / T_{\text{sel}})$ ▷ Hard top- k selection
 - 6: $G \leftarrow \text{softmax}(\mathbf{A}_{\mathcal{K}})$ ▷ Gating from unbiased affinities
 - 7: $\pi \leftarrow \text{softmax}(\mathbf{\Psi} / (T_{\text{sel}} T_{\text{ec}})); \mathbf{E} \leftarrow k \cdot \text{colsum}(\pi)$ ▷ Soft expected counts
- (Trainer executes primary task optimization step here)*

— **Dual Update Pass (Gradient-Free)** —

- 8: $t \leftarrow t + 1$
 - 9: $C_0 \leftarrow Bk/E_r; \hat{\mathbf{e}} \leftarrow (1 - \alpha)\hat{\mathbf{e}} + \alpha(\mathbf{E} - C_0)/(C_0 + \epsilon)$ ▷ Load error EMA
 - 10: $\eta_t \leftarrow \min(t/T_w, 1) \cdot \eta^*$ ▷ Linear dual learning rate warmup
 - 11: **if** $\eta_t > 0$ **then**
 - 12: $C \leftarrow C_0 c_f; \mathbf{g}^{\text{raw}} \leftarrow (\mathbf{E} - C)/(C + \epsilon)$ ▷ Instantaneous violation
 - 13: $\mathbf{g}^{\text{eff}} \leftarrow \text{DZ}_{\epsilon_s}(\mathbf{g}^{\text{raw}})$ ▷ ϵ -insensitive dead-band
 - 14: $\mathbf{u} \leftarrow m\mathbf{u} + (1 - m)\mathbf{g}^{\text{eff}}$ ▷ EMA momentum
 - 15: $\mathbf{v} \leftarrow \mathbf{v} + \eta_t \mathbf{u}$ ▷ Dual ascent update
 - 16: $\mathbf{v} \leftarrow \mathbf{v} - \text{mean}(\mathbf{v})$ ▷ Gauge-fixing mean shift
 - 17: $\mathbf{v} \leftarrow \text{clamp}(\mathbf{v}, -v_{\text{max}}, v_{\text{max}})$ ▷ Symmetric box clamping
 - 18: **end if**
 - 19: $\mathbf{W} \leftarrow \text{normalize}(\mathbf{W})$ ▷ Row-wise unit sphere projection
 - 20: **return** gates G , indices $\mathcal{K}$, expected counts $\mathbf{E}$
-

B Algorithmic Specifications: Proportional Bias Routing

Algorithm 2 Proportional Bias Routing Step

- 1: **Persistent state:** centroids $\mathbf{W}$; bias $\mathbf{b}$; sensor $\hat{\mathbf{e}}$; counter t
- 2: **Hyperparameters:** $T_{\text{sel}}, T_{\text{ec}}, \gamma, \alpha$

— **Primal Forward Pass** —

- 3: $\mathbf{A} \leftarrow \mathbf{W}\mathbf{X}^\top / \sqrt{d_\ell}$ ▷ Scaled cosine affinities
- 4: $\mathbf{\Psi} \leftarrow \mathbf{A} + \mathbf{b}$ ▷ Bias-adjusted control law
- 5: $\mathcal{K} \leftarrow \text{Top-}k(\mathbf{\Psi}/T_{\text{sel}})$ ▷ Hard top- k selection
- 6: $G \leftarrow \text{normalize}(\mathbf{A}_{\mathcal{K}})$ ▷ Decoupled proportional gate ($1/k$ fallback)
- 7: $\pi \leftarrow \text{softmax}(\mathbf{\Psi}/(T_{\text{sel}}T_{\text{ec}})); \quad \mathbf{E} \leftarrow k \cdot \text{colsum}(\pi)$ ▷ Soft expected counts
(Trainer executes primary task optimization step here; hard counts $\mathbf{c}$ are tallied from $\mathcal{K}$)

— **Bias Update Pass (Gradient-Free)** —

- 8: $t \leftarrow t + 1$
 - 9: $C_0 \leftarrow Bk/E_r; \quad \hat{\mathbf{e}} \leftarrow (1 - \alpha)\hat{\mathbf{e}} + \alpha(\mathbf{E} - C_0)/(C_0 + \varepsilon)$ ▷ Load error EMA
 - 10: $\bar{c} \leftarrow \frac{1}{E_r} \sum_j c_j; \quad \mathbf{g}^{\text{raw}} \leftarrow (\mathbf{c} - \bar{c})/(\bar{c} + \varepsilon)$ ▷ Hard-count violation
 - 11: $\mathbf{b} \leftarrow \mathbf{b} - \gamma \mathbf{g}^{\text{raw}}$ ▷ Proportional bias update (mean-preserving)
 - 12: $\mathbf{W} \leftarrow \text{normalize}(\mathbf{W})$ ▷ Row-wise unit sphere projection
 - 13: **return** gates G , indices $\mathcal{K}$, expected counts $\mathbf{E}$
-

C Theoretical Analysis and Proofs (Control Law)

This appendix provides a rigorous mathematical formalization of Stochastic Augmented Lagrangian Control (SLC) for Mixture-of-Experts (MoE) routing. We contrast the distributional optimization of SLC against DeepSeek-style penalty methods, formulate the stochastic augmented Lagrangian with Exponential Moving Average (EMA) estimators, and prove the strong convexity, KKT stationarity, and stability advantages of the proposed control law.

C.1 The Stochastic MoE Resource Allocation Program

Let $\mathbf{x} \in \mathbb{R}^{d_\ell}$ be a random token representation drawn from the underlying data distribution $\mathcal{D}$. In a sparsely-gated Mixture-of-Experts layer with E_r routed experts and routing sparsity k , the router computes token-expert affinities A_{ij} and assigns decision variables $P_{ij} \in [0, 1]$.

Instead of enforcing hard capacity constraints on every isolated mini-batch (which introduces high-frequency combinatorial noise), stochastic optimization targets distributional equilibrium over the infinite horizon, aligning with standard principles of resource allocation under uncertainty [4]:

$$\begin{aligned} \text{maximize} \quad & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\sum_{i=1}^N \sum_{j=1}^{E_r} P_{ij} A_{ij} \right] \end{aligned} \quad (28)$$

$$\text{subject to} \quad \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\sum_{i=1}^N P_{ij} \right] = C_0, \quad \forall j \in \{1, \dots, E_r\}, \quad (29)$$

where $C_0 = \frac{Bk}{E_r}$ represents the per-expert target capacity for a mini-batch of size B .

C.2 The Stochastic Augmented Lagrangian and EMA Estimation

Because computing the true expectation $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}$ across the entire data manifold is intractable during gradient-based training, we employ an Exponential Moving Average (EMA) as a recursive temporal filter to approximate expected mini-batch loads.

The Stochastic Augmented Lagrangian $\mathcal{L}(\mathbf{P}, \mathbf{W}, \mathbf{v})$ at training iteration t is formulated as:

$$\mathcal{L}(\mathbf{P}, \mathbf{W}, \mathbf{v}) = \hat{\mathbb{E}}_t \left[\sum_{i=1}^N \sum_{j=1}^{E_r} P_{ij} A_{ij} \right] - \underbrace{\sum_{j=1}^{E_r} v_j \left(\hat{E}_{t,j} - C_0 \right)}_{\text{Expected Violation } \hat{e}_{t,j}} - \underbrace{\frac{\rho}{2} \sum_{j=1}^{E_r} \left(\hat{E}_{t,j} - C_0 \right)^2}_{\text{Augmentation Penalty}} \quad (30)$$

where $\hat{E}_{t,j}$ is the soft expected load proxy defined in (4), $\hat{e}_{t,j}$ is the smoothed load-error sensor, $\mathbf{v} \in \mathbb{R}^{E_r}$ is the dual variable vector representing expert shadow prices, and $\rho > 0$ is the quadratic augmentation coefficient.

C.3 Comparative Philosophy: Lagrangian Control vs. Heuristic Bias Updates

To understand why SLC achieves superior stability, we examine the structural differences between optimization-driven control and auxiliary-loss-free heuristics:

- **Stochastic Augmented Lagrangian Control (SLC):** Formulates routing as a constrained optimization problem solved via adaptive dual control. Persistent dual prices $\mathbf{v}$ and quadratic augmentations ρ work in tandem to enforce strict capacity limits, providing provable saddle-point stability and local mini-batch smoothing (as detailed in Section 3.3).

- **DeepSeek Proportional Bias Routing:** Relies on a heuristic integral controller driven directly by hard token dispatch counts (Section 3.4). While effective at scale due to its simplicity, it operates as an unconstrained feedback loop without a formal Lagrangian foundation, making its stability dependent entirely on manual learning-rate tuning (γ) rather than provable saddle-point convergence.

C.4 Theorem: Convexity and Stationarity of the Routing Affinity

Theorem 1 (Strong Convexity and Optimal Routing Affinity). *For any positive augmentation parameter $\rho > 0$, the stochastic augmented Lagrangian functional $\mathcal{L}$ is locally strongly convex with respect to the routing assignment space, admitting a unique local minimizer per step. The resulting KKT stationarity condition yields the duality-adjusted routing control signal:*

$$\Psi_{ij}^{SLC} = A_{ij} - v_j - \rho \hat{e}_{t,j}. \quad (31)$$

Proof. Consider the objective with respect to the routing assignment parameters. Taking the functional derivative of $\mathcal{L}$ with respect to the routing probability assignment P_{ij} :

$$\frac{\partial \mathcal{L}}{\partial P_{ij}} = A_{ij} - v_j \frac{\partial \hat{E}_{t,j}}{\partial P_{ij}} - \rho (\hat{E}_{t,j} - C_0) \frac{\partial \hat{E}_{t,j}}{\partial P_{ij}}. \quad (32)$$

Recognizing that the marginal sensitivity of the expected load tensor with respect to an individual routing weight is $\frac{\partial \hat{E}_{t,j}}{\partial P_{ij}} = 1$, and substituting the smoothed EMA sensor $\hat{e}_{t,j} = \hat{E}_{t,j} - C_0$, we obtain:

$$\frac{\partial \mathcal{L}}{\partial P_{ij}} = A_{ij} - v_j - \rho \hat{e}_{t,j}. \quad (33)$$

Setting the stationarity condition to zero and isolating the terms driving the pre-softmax selection distribution yields the closed-loop routing control signal (31).

Furthermore, the addition of the quadratic augmentation term $\frac{\rho}{2} \sum_j (\hat{E}_{t,j} - C_0)^2$ injects positive curvature into the objective’s Hessian matrix with respect to the load subspace, ensuring that $\lambda_{\min}(\nabla_{\mathbf{P}}^2 \mathcal{L}) \geq \rho > 0$. This guarantees strict local convexity and numerical conditioning of the routing objective. $\square$

C.5 Proposition: Dual Ascent Convergence under Temporal Filtering

Proposition 1 (Dual Update Stability). *Let the shadow prices evolve via dual ascent driven by the smoothed constraint violation:*

$$\mathbf{v}_{t+1} = \mathbf{v}_t + \eta_t \hat{\mathbf{e}}_t, \quad (34)$$

where $\eta_t > 0$ is the dual learning rate and $\hat{\mathbf{e}}_t$ represents the filtered load-error vector. Under standard multi-timescale separation where dual updates evolve slower than primal parameters, the filtered dual sequence tracks the constraint manifold, ensuring that the expected constraint violation converges to zero ($\lim_{t \rightarrow \infty} \mathbb{E}[\hat{\mathbf{e}}_t] = \mathbf{0}$) at a stable saddle-point equilibrium [2].

Proof. Because the Lagrangian functional is locally strongly convex in the primal space (Theorem 1) due to the quadratic augmentation term $\rho > 0$, the dual objective function is smooth and well-behaved. The dual update functions as a discrete-time integral controller.

The Exponential Moving Average (EMA) filter $\hat{\mathbf{e}}_t$ acts as a recursive low-pass filter that attenuates high-frequency mini-batch sampling noise. Consequently, the effective update direction $\eta_t \hat{\mathbf{e}}_t$ provides a reliable gradient ascent trajectory on the dual surface. Under standard stochastic approximation and multi-timescale convergence guarantees for augmented Lagrangians, the filtered dual sequence contracts toward the optimal multiplier vector $\mathbf{v}^*$, guaranteeing asymptotic feasibility and zero steady-state load error. $\square$

D Theoretical Foundation of Hard Top- K Primal Routing

A pervasive misconception in deep learning router design is that discrete Top- k selection is an ad-hoc, non-differentiable heuristic necessitating continuous relaxations such as softmax. Here, we demonstrate that when framed through convex optimization and network flow theory, the hard Top- k operator over the duality-adjusted affinities is the exact, closed-form solution to a linear program, incurring a zero integrality gap.

D.1 The Hypersimplex Linear Program Formulation

Let token-to-expert dispatch be formulated as a continuous optimization problem for a token i . Whether utilizing the duality-adjusted control signal from our Lagrangian framework ($\Psi_{ij}^{\text{SLC}} = A_{ij} - v_j - \rho \hat{e}_{t,j}$) or the heuristic selection bias from the DeepSeek baseline ($\Psi_{ij}^{\text{bias}} = A_{ij} + b_j$), any additive adjustment yields a net expert utility score Ψ_{ij} .

The continuous routing assignment vector $\mathbf{p}_i = [P_{i1}, \dots, P_{iE_r}]^\top$ maximizes linear utility over the (E_r, k) -hypersimplex polytope, denoted as $\Delta(E_r, k)$ (defined as the convex hull of all binary vectors in $\{0, 1\}^{E_r}$ whose entries sum to k):

$$\text{maximize} \quad \sum_{j=1}^{E_r} P_{ij} \Psi_{ij} \quad (35)$$

$$\text{subject to} \quad \sum_{j=1}^{E_r} P_{ij} = k, \quad (36)$$

$$0 \leq P_{ij} \leq 1, \quad \forall j \in \{1, \dots, E_r\}. \quad (37)$$

This feasible region corresponds precisely to the uniform matroid base polytope of rank k [1].

D.2 Total Unimodularity and Matroid Optimality

By casting the constraint matrix of the hypersimplex in standard form, it can be shown to possess the total unimodularity property. Because the constraint matrix is totally unimodular and the boundary vector is integral, every extreme point (vertex) of the continuous polyhedron $\Delta(E_r, k)$ is strictly binary [1]:

$$\text{ext}(\Delta(E_r, k)) = \left\{ \mathbf{p} \in \{0, 1\}^{E_r} : \sum_{j=1}^{E_r} p_j = k \right\}. \quad (38)$$

Consequently, the continuous relaxation has a zero integrality gap. Furthermore, maximizing a linear objective over a uniform matroid base polytope is solved globally and exactly by the greedy algorithm. Sorting the affinities Ψ_i in descending order yields the exact binary vertex via the indicator condition:

$$P_{ij}^* = \begin{cases} 1, & j \in \mathcal{K}_i, \\ 0, & \text{otherwise,} \end{cases} \quad (39)$$

where $\mathcal{K}_i = \text{Top-}k_j(\Psi_{ij})$.

D.3 Implications for Router Design

This polyhedral exactness explains why auxiliary-loss-free routers utilizing hard Top- k achieve superior stability over continuous relaxations. Standard approximations (such as softmax over the expert set) smooth out discrete choices, introducing approximation slack and sensitivity into a problem that is analytically solvable as an exact linear program. Hard Top- k selection bypasses these approximations, directly executing the optimal extreme-point solution of the routing polytope.

E Implementation and Hyperparameter Configurations

This appendix provides the full configuration specifications, dataset details, and hyperparameter settings utilized for the pre-training and evaluation experiments in Section 4.

E.1 Dataset and Tokenization Details

Experiments utilize the TinyStories corpus ([roneneldan/TinyStories](#)) [8], comprising synthetic, narrative-focused short stories generated via GPT-3.5 and GPT-4 under restricted child-like vocabulary and syntax constraints. Text sequences are tokenized using the standard GPT-2 byte-pair encoding (BPE) tokenizer, establishing a fixed vocabulary size of $|\mathcal{V}| = 50,257$ tokens [14].

E.2 Transformer, MLA, and MoE Architecture Specifications

The model is structured as a fine-grained Mixture-of-Experts (MoE) GPT architecture integrated with Multi-Head Latent Attention (MLA) compression layers with a batch size of 16. The complete architectural hyperparameter profile is detailed in Table 3.

Table 3: Full architectural and optimization hyperparameter specifications.

Category	Parameter	Value
Transformer	Layers (N_{layer})	6
	Embedding dim (d_{model})	512
	Attention heads (N_{head})	8
	Context window (T)	256
MLA	Compression ratio	0.5
	KV latent dim (d_{latent})	128
	RoPE dim (d_{rope})	16
MoE	Routed experts (E_r)	16
	Expert hidden dim (h_r)	512
	Routing sparsity (k)	2
	Shared experts	2
	Shared hidden dim	1,024
Optimization	Iterations	30,000
	Batch size (B)	16
	Warm-up steps	200

E.3 Training Procedure and Hardware Environment

Optimization is driven by an adaptive first-order optimizer paired with a cosine learning-rate decay schedule featuring linear warm-up. The primal learning rate increases linearly from 0 to a peak of $\eta = 3 \times 10^{-4}$ over the initial 200 optimization steps, followed by cosine annealing down to a minimum floor of 3×10^{-5} .

All models are trained under mixed-precision execution (`bfloat16/float16`) with dynamic gradient scaling. Training is executed locally on a single consumer workstation equipped with an NVIDIA RTX 3060 (12GB VRAM) GPU.

F Unified Mathematical Notation

Symbol	Domain / Type	Mathematical Meaning
B	$\mathbb{N}^+$	Mini-batch size; parallel sequence streams processed.
T	$\mathbb{N}^+$	Active sequence length per batch element.
N	$\mathbb{N}^+$	Total active token count in a mini-batch ($N = B \cdot T$).
E_r	$\mathbb{N}^+$	Number of conditionally routed Mixture-of-Experts.
k	$\mathbb{N}^+$	Routing sparsity; top- k experts selected per token.
d_ℓ	$\mathbb{N}^+$	Latent feature dimension for routing and attention.
$\mathbf{X}$	$\mathbb{R}^{N \times d_\ell}$	Input latent token representation matrix.
$\mathbf{W}$	$\mathbb{R}^{E_r \times d_\ell}$	Unit-sphere normalized expert centroid matrix ($\ \mathbf{w}_j\ _2 = 1$).
$\mathbf{A}$	$\mathbb{R}^{N \times E_r}$	Scaled cosine affinity matrix ($A_{ij} = \mathbf{x}_i^\top \mathbf{w}_j / \sqrt{d_\ell}$).
$T_{\text{sel}}, T_{\text{ec}}$	$\mathbb{R}^+, \mathbb{R}^+$	Selection and expected-count proxy temperatures.
$\boldsymbol{\pi}$	$\mathbb{R}^{N \times E_r}$	Soft expected routing probability distribution.
$\mathbf{E}$	$\mathbb{R}^{E_r}$	Soft expected token load counts across experts.
$\mathbf{c}$	$\mathbb{N}_0^{E_r}$	Hard token dispatch counts tallied from selections.
G	Matrix / Tensor	Final routed expert gating weights.
$\mathbf{v}$	$\mathbb{R}^{E_r}$	Dual shadow price vector (expert congestion tax).
$\hat{\mathbf{e}}$	$\mathbb{R}^{E_r}$	Recursive load-error estimator tracking relative allocation error.
ρ	$\mathbb{R}^+$	Augmented Lagrangian quadratic penalty parameter.
η^*	$\mathbb{R}^+$	Target dual learning rate asymptote.
m	$[0, 1)$	Dual momentum decay coefficient.
ϵ_s	$\mathbb{R}^+$	ϵ -insensitive dead-band slack half-width parameter.
C_0, C, c_f	$\mathbb{R}^+, \mathbb{R}^+, \mathbb{R}^+$	Target expert capacity ($C_0 = B k / E_r$), capacity-scaled target ($C = C_0 c_f$), and capacity factor scaling coefficient.
$\mathbf{b}$	$\mathbb{R}^{E_r}$	Additive per-expert selection bias vector.
γ	$\mathbb{R}^+$	Proportional bias update step-size gain.
T_w	$\mathbb{N}^+$	Warm-up iteration duration for the dual learning rate.

Table 5: Master symbol reference for MoE routing architectures and control dynamics.